PUBLISHED PAPER · ZHAO, CRANE & BEZBRADICA · 2026

Paper 1 — MSPCIFormer

Multi-Scale Patching Channel-Independent Transformer for Cryptocurrency Price Forecasting

> **Research question.** Can a Transformer forecast volatile cryptocurrency prices more effectively by learning **short- and long-range patterns together** while avoiding noisy, unstable relationships across assets? **Answer proposed:** multi-scale temporal patching + channel-independent processing + learned scale aggregation.

1 / Problem and motivation

Crypto markets change behavior across time. A few days may contain rapid trading-driven moves; several weeks may reveal trends; longer periods may show new market regimes. A useful forecasting model must recognize several **temporal scales** simultaneously.

Where existing approaches struggle

- **ARIMA:** primarily models linear time-series structure.
- **GARCH:** primarily targets conditional volatility rather than rich price relationships.
- **LSTM:** sequential recurrence can make long dependencies harder to learn efficiently.
- **Standard Transformers:** attention can be costly and sensitive to noisy observations.
- **Single-size patches:** one resolution may miss patterns visible at other time scales.

2 / How MSPCIFormer works

1. **Patch at multiple scales.** Group historical observations into different-length segments (small = local changes; large = broader trends).
2. **Process channels independently.** Apply a shared temporal model to each cryptocurrency price series separately, limiting reliance on changing cross-asset correlations.
3. **Use encoder-only self-attention.** Identify which historical patches are relevant to a forecast, including non-adjacent patterns.
4. **Aggregate scales with learned weights.** Combine scale-specific representations/predictions rather than treating every temporal resolution as equally informative.

Intuition: one scale may detect a two-day reversal while another detects a persistent month-long trend; the final model uses information from both.

3 / Data and experimental setup

1,728 daily observations | 10 Apr 2020–1 Jan 2025

Inputs: ADA, BNB, BTC, DOGE, ETH, SOL, TRX and XRP daily closing prices from Yahoo Finance. **Forecasted assets:** BTC, ETH and ADA.

Historical input: reported 5-day lookback. **Forecast horizons:** 1, 3, 5, 7 and 15 days. **Validation:** chronological ~70/10/20 train/validation/test split plus five sequential expanding-window walk-forward folds.

4 / Metrics: what success means

Metric	Interpretation
MAE / RMSE	Size of forecast errors; lower is better (RMSE penalizes large misses more).
MSE / MAPE	Squared error / relative percentage error; lower is better.
MDA	Correct direction of price change; higher is better.
Sharpe	Excess trading return per unit of return volatility; higher is better.
Max drawdown	Largest peak-to-trough portfolio loss; smaller magnitude is better.

Representative 1-day forecast results

Asset / test	MAE ↓	MDA ↑	Sharpe ↑
BTC holdout	0.08909	53.04%	1.180
BTC walk-fwd.	0.04023	54.35%	0.654
ETH walk-fwd.	0.03606	53.77%	0.823

MAE uses the paper's reported target scale, **not dollars or percentages**. Holdout and walk-forward are different evaluation protocols; comparing their MAE values does not establish improvement.

5 / Findings, efficiency and limits

269,858	2.51 ms	20.73 MB
model parameters	per-sample latency*	peak memory*

*Walk-forward averages across the paper's assets and forecast horizons on its test setup; not general hardware benchmarks.

- **Architectural result:** multi-scale, channel-independent modeling was competitive and often strong relative to the tested Transformer baselines (DLinear, LSTM, N-BEATS, TimesNet, MSGNet, iTransformer, TimeXer).
- **Directional skill remains modest:** representative MDA values are roughly 53–54%, not near-perfect trend prediction.
- **Accuracy is not profit:** DLinear achieved about **2.08 Sharpe** on BTC 1-day holdout, above MSPCIFormer's 1.18.

DEVELOPER TAKEAWAY

Paper 1 contributes the temporal architecture: learn short/medium/long price patterns efficiently, separately for each channel, then learn which scales matter for a forecast. It does *not* prove that lower price error guarantees better trades.

Publication: doi:10.1111/exsy.70343

PUBLISHED PAPER · GURGUL, LESSMANN & HÄRDLE · 2025

Paper 2 — Deep Learning + NLP

Deep Learning and NLP in Cryptocurrency Forecasting: Integrating Financial, Blockchain, and Social Media Data

> **Research question.** Do market prices alone tell us enough about future crypto movements, or can **blockchain activity and news/social sentiment** improve forecast quality and simulated trading profitability? The paper tests both the additional information and **which forecasting targets are useful to trade**.

1 / Research gap and novelty

- **Price-only models** cannot directly see public narratives or on-chain network activity.
- **Lexicon sentiment (VADER)** can miss context and whether a statement is financially bullish/bearish.
- **Daily signals** can be noisy and lead to excessive transactions.

Distinctive contribution: a systematic comparison of multi-source features and NLP extractors, particularly **zero-shot BART-MNLI financial hypotheses**, alongside forecasts of **local price maxima and minima**. It is an empirical framework, not a newly invented Transformer architecture.

2 / Data: what is fed into models?

Assets: BTC and ETH; histories extend to March 2023 (BTC from Aug 2011; ETH from Aug 2015). Information sources include:

- **Market:** prices, returns, exchange volume, volatility, 48 technical indicators, macro variables.
- **On-chain:** addresses, balances, transactions, hashrate and block statistics.
- **Text:** ~1.9M tweets, 338K Reddit threads, 55K news headlines.
- **Other:** GitHub activity, Google Trends, engagement measures; feature lags up to 14 days.

3 / Methodology: text → prediction

1. **Extract sentiment signals.** Turn articles/posts into quantitative features using several NLP methods.
2. **Join features by time.** Combine daily sentiment aggregates with contemporaneously available market, technical and blockchain metrics.
3. **Fit forecasting models.** Compare regression / logistic baselines, XGBoost, MLP (FNN), LSTM and Temporal Fusion Transformer (TFT).
4. **Evaluate forecast + trading.** Use seven time-series validation folds and simulated trading strategies, not just classification accuracy.

How the NLP alternatives differ

- **VADER:** dictionary-based positive/negative tone.
- **Twitter-RoBERTa:** contextual social-media sentiment.
- **BART-large-MNLI:** zero-shot score for hypotheses such as “this news is bullish for BTC.”
- **Fine-tuned RoBERTa:** adapts language model to price-movement labels; fine-tuning is not automatically superior.

4 / Targets: what gets predicted?

Daily returns / direction: tomorrow's price change, or up versus down. **Local extrema:** whether a date is a local *minimum* (lower than surrounding prices) or *maximum* (higher), defined with ±7-, ±14- and ±21-day windows.

> **Why extrema?** A strategy can try to act near meaningful bottoms/tops instead of trading on every daily fluctuation. Historical extrema labels require future prices to become known; **future values cannot be model inputs at prediction time**.

5 / Results: what did the study find?

Reported averages across BTC/ETH classification targets and seven validation folds:

Model	Accuracy	AUC-ROC	Excess profit*
OLS / Logit	80.25%	0.6782	+67.48 pp
XGBoost	80.57%	**0.6998**	+126.12 pp
MLP / FNN	**80.65%**	**0.6797**	**+138.61 pp**
LSTM	80.28%	0.6526	+83.88 pp
TFT	79.71%	0.5653	+11.13 pp

*Percentage points above simulated buy-and-hold, **not** guaranteed return. The headline ~80% accuracy is **not** 80% accuracy on ordinary next-day price direction; it averages multiple, sometimes class-imbalanced tasks.

XGBoost feature importance by source

Category	BTC	ETH
Technical indicators	37.22%	56.91%
NLP features	12.73%	22.20%
Transaction / balance	24.62%	—

Feature gain is not a causal contribution or the accuracy increase from adding sentiment.

6 / Interpretation and limits

- **MLP led** aggregated accuracy and simulated excess profit; **XGBoost led AUC-ROC**. Bigger models were not automatically better.
- Sentiment and blockchain variables carried useful predictive information in the studied setups, but value varies across assets and targets.
- Main trading simulation excluded transaction costs; reported backtest results need realistic cost and out-of-sample scrutiny.

DEVELOPER TAKEAWAY

Paper 2 contributes information and target design: augment price features with on-chain + NLP signals, test financially meaningful text hypotheses, and consider market turning points rather than only daily direction.

Publication: doi:10.1016/j.ijforecast.2025.02.007

RESFINANCE ARCHITECTURE + REPORTED PROJECT EVIDENCE · NOVEMBER 2026

Combining both papers in ResFinance

From multi-source financial data to adaptive fusion and multi-scale crypto return forecasts

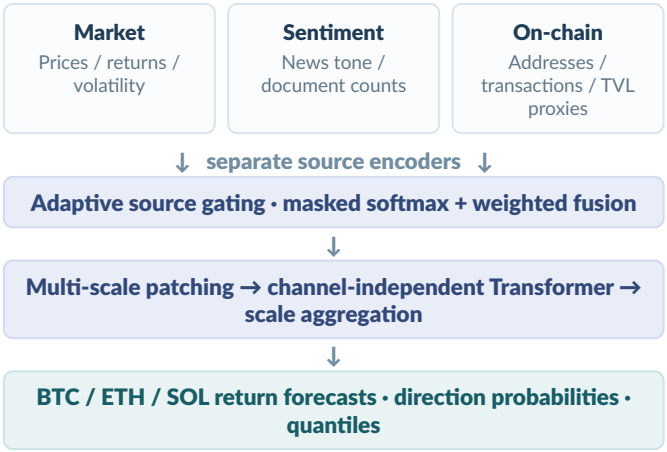
> **Central hypothesis.** A multi-scale Transformer that **learns how much to trust each available data source** (market, sentiment, on-chain) could outperform market-only or fixed-weight fusion. **This is a research hypothesis, not yet a demonstrated predictive advantage.**

1 / What comes from each paper?

Origin	What ResFinance uses
Paper 1	Multi-scale patches, channel-independent Transformer, learnable aggregation across temporal scales.
Paper 2	Motivation for combining market, sentiment and blockchain sources; rigorous financial evaluation.
ResFinance	**Adaptive Multi-Source Gating**: learned source weights before temporal forecasting.

The system **adapts** both works; it does not exactly reproduce their datasets, NLP pipelines, model architectures, or trading experiments.

2 / End-to-end architecture



Where the proposed novelty sits

Source weighting chooses the contribution of market vs. sentiment vs. on-chain features. **Scale weighting** (Paper 1) separately combines short vs. longer temporal patterns. Both sets of weights are learned; they are not interchangeable.

3 / What adaptive gating calculates

For available sources: $w_i = \exp(z_i) / \Sigma_j \exp(z_j)$

Fused representation: $h_{fused} = w_m h_m + w_s h_s + w_c h_c$

Here, each h is a learned source embedding and w is its importance weight; available weights sum to 1. An unavailable source is **masked to zero weight**. Unlike fixed-equal fusion (about 33/33/33), weights can vary by asset and time.

4 / Current implementation

Source providers: Coin Metrics (BTC/ETH/SOL market and BTC/ETH on-chain), GDELT (news aggregates), DefiLlama (SOL proxies). The training data is **not** the full Twitter/Reddit/BART corpus used by Paper 2.

Input: 16-day lookback; 3 sources with 3 feature slots/source. **Model:** source encoders + learned gating, patch lengths **2 / 4 / 8**, channel-independent Transformer, learned scale aggregation.

Outputs: predicted future returns at **1, 3, 5, 7 and 15 days**, direction probability and 10th/50th/90th quantiles (followed by interval calibration). Unlike Paper 1, the primary output is future *return*, not raw price.

Model implementation: **ml/models/mspciformer.py** · class **ThreeAssetMSPCIFormer**.

5 / How to test the research claim

- **Baselines:** naive zero-return, Ridge, LSTM and market-only model.
- **Fusion comparison:** fixed-equal fusion versus learned adaptive gating.
- **Source ablations:** remove market, sentiment, or on-chain features one at a time.
- **Protocol:** chronological / expanding walk-forward evaluation with an embargo to avoid look-ahead leakage; track MAE, direction, uncertainty coverage and financial outcomes.

Recorded research outcomes

Experiment	Adaptive MAE	Naive MAE
Original provider-backed	0.06636	0.06367
Forward holdout*	0.05297	0.05297

Original model: **82.40%** interval coverage but worse MAE than naive → rejected. *Forward holdout: **88.05%** coverage; validation selected reliability coefficient **c = 0**, shrinking return forecasts to zero. It passed the existing promotion rule on equality, **not on predictive improvement**.

6 / Honest conclusion + next evidence

Implemented: architecture, data pipeline, baselines/ablations, evaluation and downstream risk-aware portfolio integration. **Not demonstrated:** that nonzero adaptive gating reliably beats simpler forecasts on new market data.

Next: fix incomplete sentiment-feature consumption, secure point-in-time coverage for each source, test consistent lift across folds/assets, then freeze and evaluate on a genuinely untouched future period.

ONE-SENTENCE SUMMARY

Paper 1 teaches the model how to learn from time; Paper 2 shows why additional information matters; ResFinance tests whether learned source importance improves forecasting before that multi-scale temporal analysis.

Project: github.com/haseebshaik00/ResFinance · Status based on 5 Nov 2026 guide.